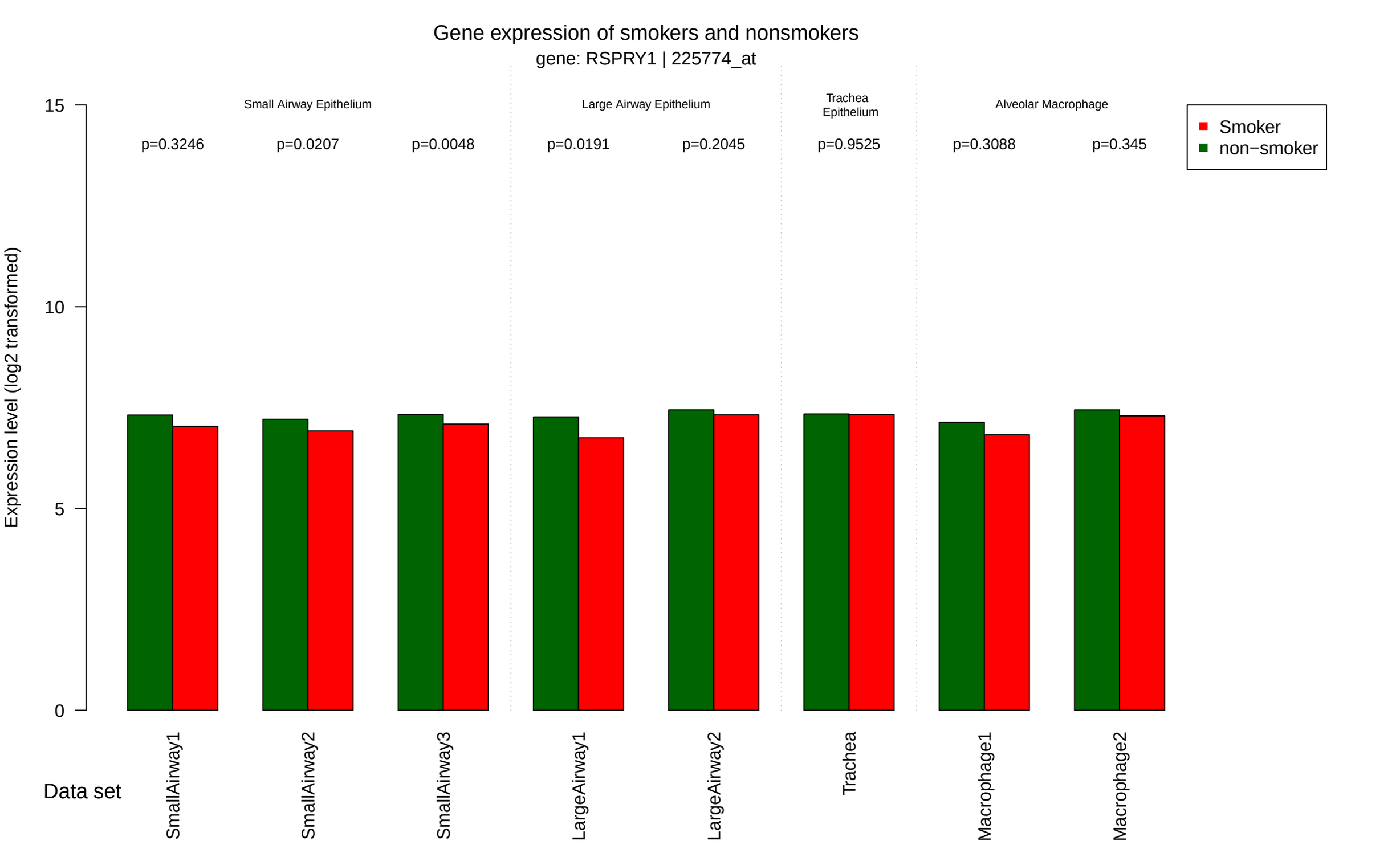
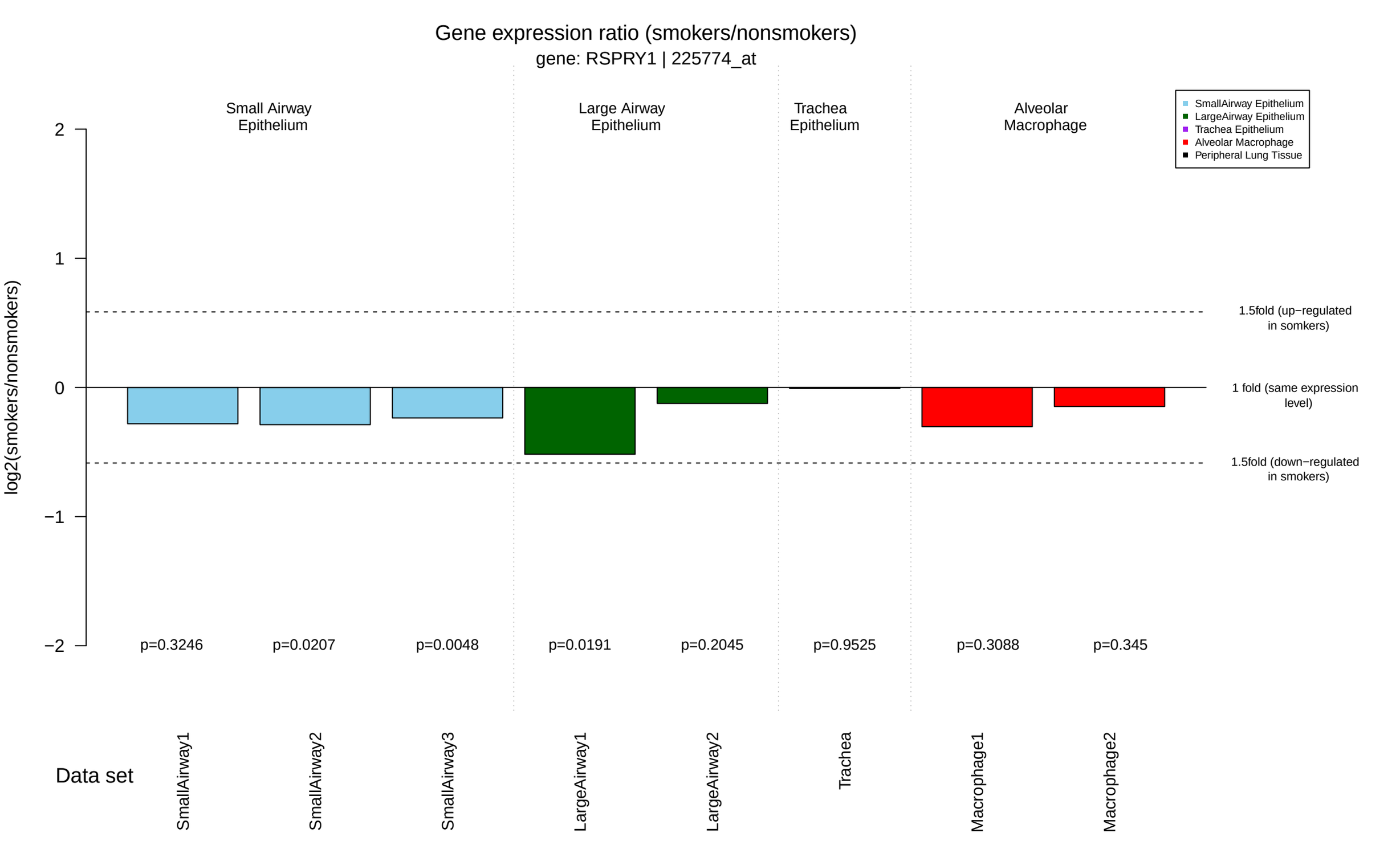
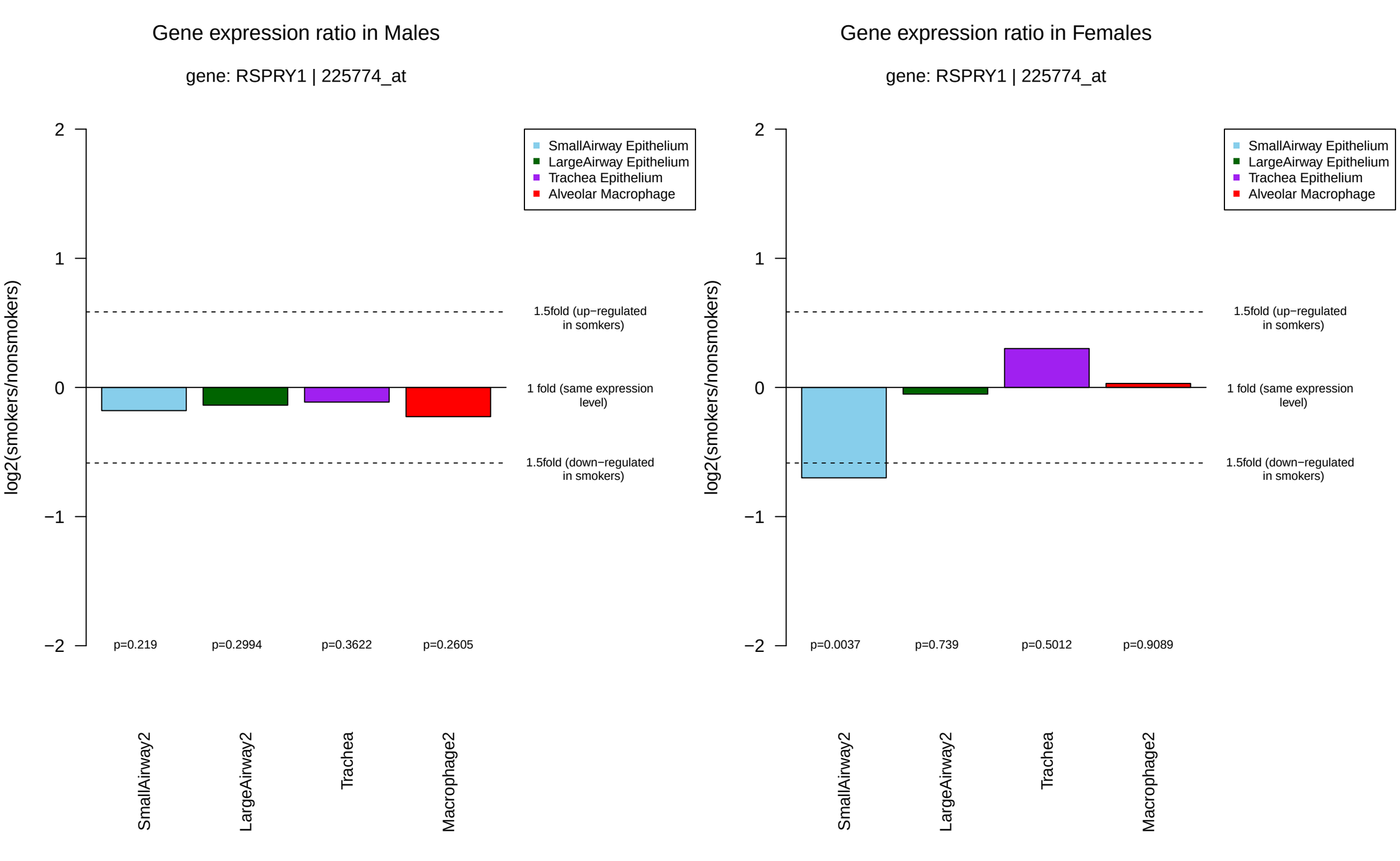
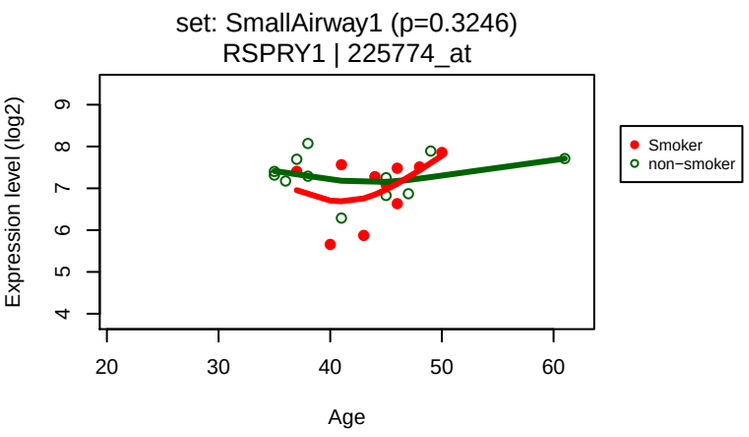
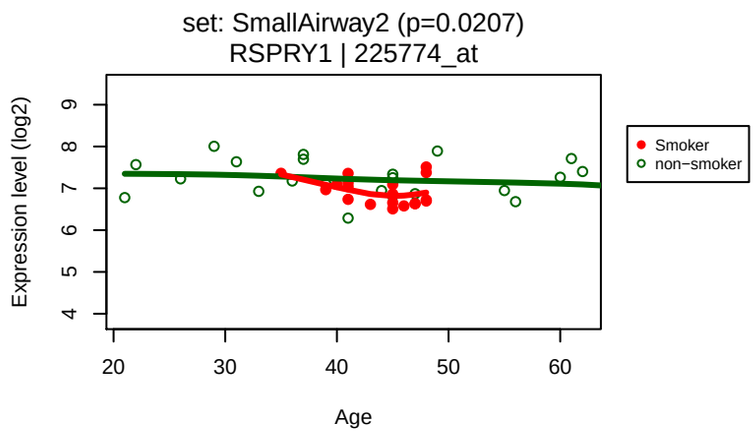
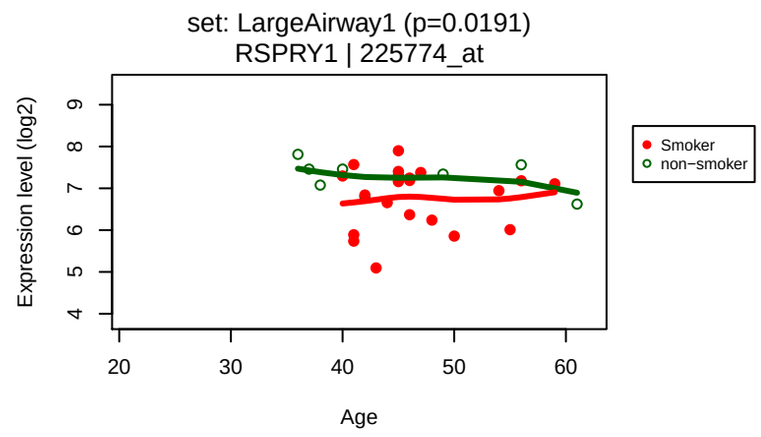

Gene expression of smokers and nonsmokers
gene: RSPRY1 | 225774_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3246
p=0.0207
p=0.0048
p=0.0191
p=0.2045
p=0.9525
p=0.3088
p=0.345
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

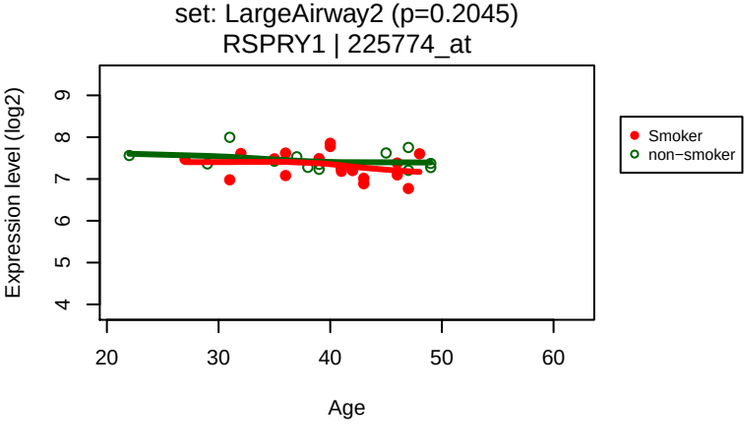

Gene expression ratio (smokers/nonsmokers)
gene: RSPRY1 | 225774_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3246
p=0.0207
p=0.0048
p=0.0191
p=0.2045
p=0.9525
p=0.3088
p=0.345
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

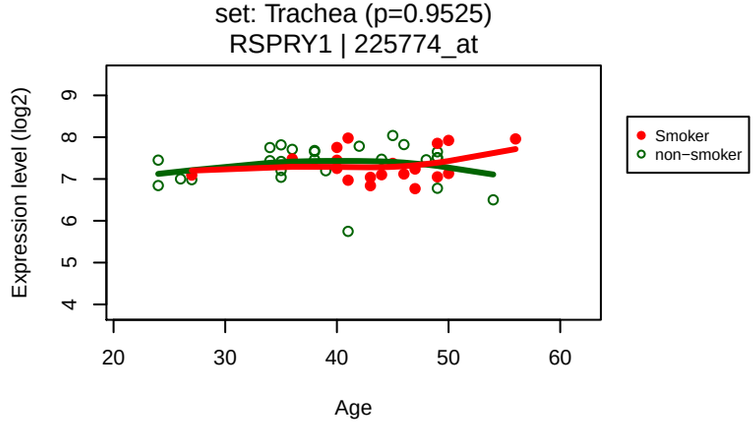

Gene expression ratio in Males
gene: RSPRY1 | 225774_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.219
p=0.2994
p=0.3622
p=0.2605
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RSPRY1 | 225774_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0037
p=0.739
p=0.5012
p=0.9089
SmallAirway2
LargeAirway2
Trachea
Macrophage2

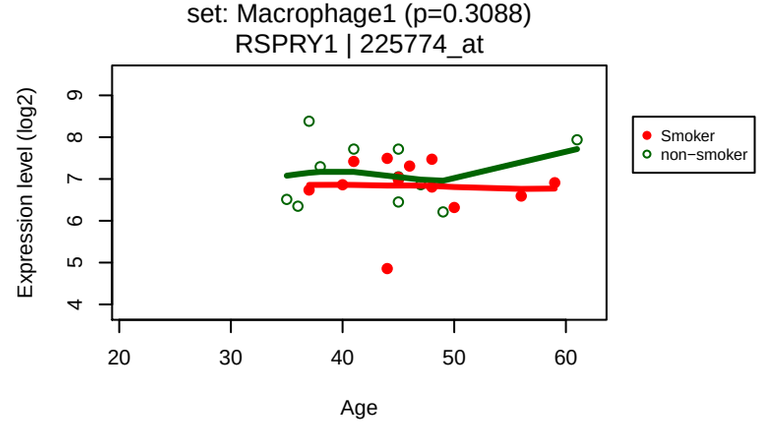

set: SmallAirway1 (p=0.3246)
RSPRY1 | 225774_at
Expression level (log2)
Age
Smoker
non−smoker

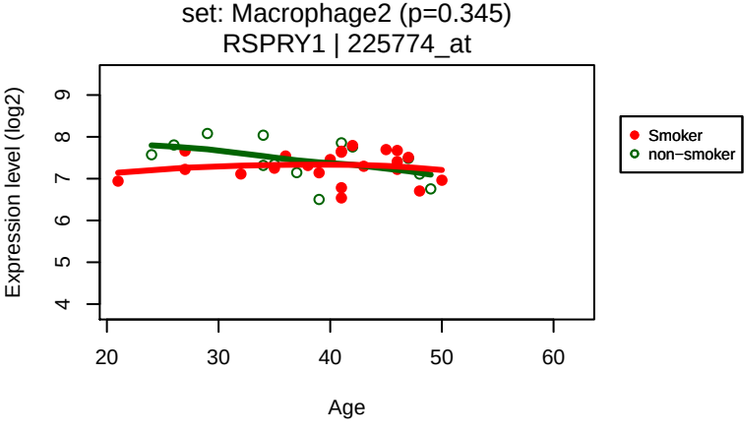

set: SmallAirway2 (p=0.0207)
RSPRY1 | 225774_at
Expression level (log2)
Age
Smoker
non−smoker

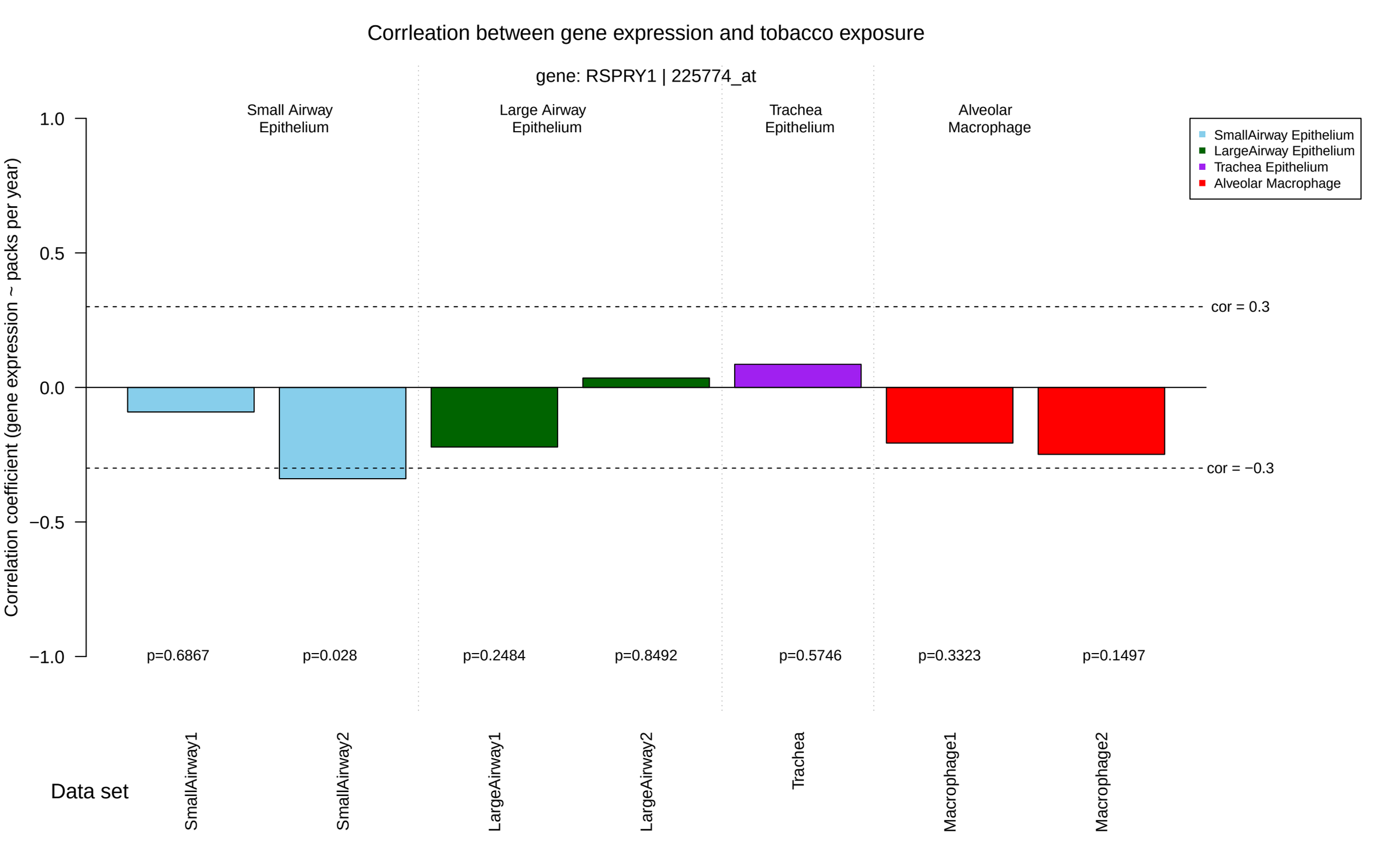

set: LargeAirway1 (p=0.0191)
RSPRY1 | 225774_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2045)
RSPRY1 | 225774_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9525)
RSPRY1 | 225774_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3088)
RSPRY1 | 225774_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.345)
RSPRY1 | 225774_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RSPRY1 | 225774_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6867
p=0.028
p=0.2484
p=0.8492
p=0.5746
p=0.3323
p=0.1497
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2